

Business Requirement Document

Product Catalog Management Module

ShopSphere E-Commerce Platform — ShopSphere Platform Modernization Program (FY2026)

Document Title	Business Requirement Document — Product Catalog Management Module
Document ID	BRD-EC-PCM-002
Version	1.0
Author	Neha Kapoor, Business Analyst — Catalog & Merchandising
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1. Document Information

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2. Executive Summary

ShopSphere Technologies Pvt. Ltd. operates the ShopSphere E-Commerce Platform, a B2C marketplace serving approximately 2 million monthly active users across India, UAE, and Southeast Asia. The Product Catalog Management (PCM) module is the authoritative system of record for all product, category, attribute, SKU/variant, and pricing data that powers the Product Listing Page (PLP), Product Detail Page (PDP), and catalog search experiences across web and mobile channels. This document defines the business requirements for modernizing PCM under the ShopSphere Platform Modernization Program (FY2026), addressing longstanding gaps in data quality, category taxonomy governance, multi-market attribute support, pricing approval controls, and catalog publish/unpublish workflows. The modernized PCM module will integrate tightly with SAP S/4HANA for product master synchronization, the Inventory Management System (IMS) for stock linkage, and Elasticsearch for real-time search indexing, enabling faster, more accurate, and more compliant catalog operations at scale.

3. Business Objective

The primary business objective of this initiative is to establish a single, governed, and scalable source of truth for product catalog data that reduces time-to-publish for new products, improves PDP data completeness and accuracy, and enables consistent multi-language and multi-currency product presentation across the India, UAE, and Southeast Asia markets. The initiative aims to reduce catalog-related customer complaints (incorrect pricing, missing specifications, mismatched images), reduce manual effort in bulk product uploads via automated ERP feed ingestion, and introduce a formal pricing approval workflow to prevent unauthorized or erroneous price changes from reaching customers. Ultimately, the module supports ShopSphere's broader Gross Merchandise Value (GMV) growth targets by ensuring merchandising teams can onboard, enrich, and publish products faster while maintaining data governance and regulatory compliance.

4. Project Scope

In Scope

- Product creation and editing, including title, description, brand, and category assignment.
- Category and attribute management, including taxonomy hierarchy and attribute-set governance.
- SKU/variant management covering size, color, and other variant dimensions.
- Bulk product import via automated ERP feed from SAP S/4HANA.

- Pricing and discount rule configuration, including approval workflow.
- Catalog search indexing into Elasticsearch for PLP/PDP consumption.
- Product lifecycle status management (draft, active, inactive, discontinued).
- Content enrichment, including images, descriptions, and technical specifications.

Out of Scope

- Supplier/vendor onboarding workflow (covered under a separate Vendor Management BRD).
- AI-based dynamic repricing engine.
- Building a standalone digital asset management (DAM) tool.
- Marketplace third-party seller catalog self-service portal.

5. Stakeholders

Name	Role	Department	Responsibility
Rajesh Mehta	VP of E-Commerce	Business	Executive Sponsor
Ananya Sharma	Product Owner, ShopSphere Platform	Product	Prioritization & sign-off
Neha Kapoor	Business Analyst, Catalog & Merchandising	Business Analysis	Requirements owner
Sarah Chen	Enterprise IT Architect	IT/Architecture	Technical feasibility
Michael Torres	QA Lead	Quality Assurance	Test strategy & sign-off
Priya Nair	UX Lead	Design	User experience
David Kim	DevOps Lead	Engineering/Infra	Deployment & monitoring
Fatima Al-Sayed	Compliance & Security Officer	Risk & Compliance	Regulatory compliance

6. Business Requirements

ID	Business Requirement
BR-001	The business requires a centralized catalog system that maintains product, category, and pricing data as a single source of truth across web and mobile channels for India, UAE, and Southeast Asia.
BR-002	The business requires automated, scheduled synchronization of the product master from SAP S/4HANA to eliminate manual re-keying of product data and reduce data entry errors.
BR-003	The business requires a governed category taxonomy so that merchandising teams can add or restructure categories without breaking existing PLP navigation or search filters.
BR-004	The business requires a pricing change approval workflow to prevent unauthorized or erroneous price and discount changes from being published to customers.
BR-005	The business requires enforced minimum data completeness standards for PDP content (images, description, specifications) before a product can be published.
BR-006	The business requires support for multi-language product descriptions and multi-currency pricing to serve customers in India, UAE, and Southeast Asia markets appropriately.
BR-007	The business requires near-real-time catalog search index updates so that newly published or updated products are discoverable within minutes.
BR-008	The business requires a defined product lifecycle (draft, active, inactive, discontinued) with controlled state transitions to avoid customers viewing unpublished or discontinued products.

7. Functional Requirements

Product & SKU Management

- **FR-001** The system shall allow authorized catalog users to create, edit, and archive a product record, including title, brand, category assignment, and base attributes.
- **FR-002** The system shall support creation of multiple SKUs/variants per product (e.g. size, color combinations), with each SKU carrying a unique, system-generated SKU code.
- **FR-003** The system shall enforce SKU uniqueness validation at the point of creation and during bulk import, rejecting duplicate SKU codes with a clear error message.
- **FR-004** The system shall link each SKU to its corresponding stock record in the Inventory Management System (IMS) to reflect Available-to-Promise (ATP) quantity on the PDP.

Category & Attribute Management

- **FR-005** The system shall provide a hierarchical category taxonomy editor supporting parent-child category relationships up to five levels deep.
- **FR-006** The system shall support attribute-set templates per category (e.g. "Apparel" requires size/color, "Electronics" requires wattage/warranty) to ensure consistent PDP structure.
- **FR-007** The system shall require Product Owner or designated taxonomy governance approval before a new top-level or second-level category is published live.

Bulk Import & ERP Synchronization

- **FR-008** The system shall support bulk product import via a scheduled feed from SAP S/4HANA, processing new and updated product master records in a defined file format (CSV/XML).
- **FR-009** The system shall validate each bulk import record against mandatory field rules and produce an exception report listing rejected records with reasons.
- **FR-010** The system shall allow reprocessing of corrected records from an exception report without requiring a full re-import of the batch.

Pricing & Discount Configuration

- **FR-011** The system shall allow authorized users to configure base price, promotional price, and discount rules per SKU and per market currency.
- **FR-012** The system shall route price changes exceeding a configurable threshold (e.g. more than 15% variance from current price) to a designated approver before the change takes effect.
- **FR-013** The system shall maintain a full audit trail of all pricing changes, including who made the change, previous value, new value, and approval status.

Content Enrichment & Catalog Publishing

- **FR-014** The system shall allow upload and association of multiple product images, a long-form description, and structured technical specifications per product.
- **FR-015** The system shall block publishing of a product to "Active" status unless mandatory content fields (minimum one image, description, category, and price) are complete.
- **FR-016** The system shall support a publish/unpublish action that immediately updates product visibility on PLP/PDP and triggers a corresponding Elasticsearch index update.
- **FR-017** The system shall support multi-language fields for product title, description, and specifications for at least English, Arabic, Hindi, and one Southeast Asian language (e.g. Bahasa Indonesia).

8. Non-Functional Requirements

Performance

- **NFR-001** Catalog search and PDP data retrieval APIs shall respond within 2 seconds at the 95th percentile under normal load.
- **NFR-002** Bulk import processing of up to 50,000 product/SKU records shall complete within 30 minutes end-to-end, including validation.
- **NFR-003** Elasticsearch index updates for a publish/unpublish action shall propagate within 5 minutes of the triggering event.

Security

- **NFR-004** The system shall comply with OWASP Top 10 secure coding practices for all catalog management interfaces and APIs.
- **NFR-005** Role-based access control shall restrict pricing configuration and category taxonomy changes to authorized roles only.
- **NFR-006** All data in transit between PCM, SAP S/4HANA, IMS, and Elasticsearch shall be encrypted using TLS 1.2 or higher; sensitive data at rest shall use AES-256 encryption.

Availability

- **NFR-007** The catalog management module and its customer-facing read APIs shall maintain 99.9% uptime measured monthly.
- **NFR-008** Planned maintenance windows shall not affect customer-facing PLP/PDP browsing availability.

Scalability

- **NFR-009** The system shall support up to 10x normal catalog browsing and search traffic during peak sale events such as Diwali and year-end sale campaigns without degradation beyond defined performance targets.
- **NFR-010** The system shall support a catalog size of at least 5 million active SKUs across all markets without performance degradation.

Reliability

- **NFR-011** The system shall support a Recovery Point Objective (RPO) of 15 minutes and Recovery Time Objective (RTO) of 1 hour for catalog data services.
- **NFR-012** Failed bulk import batches shall be automatically retried up to 3 times before being flagged for manual intervention.

9. User Stories

- **US-001** As a Catalog & Merchandising analyst, I want to create a new product with multiple SKU variants, so that customers can select the correct size and color on the PDP.
- **US-002** As a Catalog & Merchandising analyst, I want to bulk import products from the SAP feed, so that I do not have to manually re-key thousands of records during a new season launch.
- **US-003** As a Pricing Manager, I want price changes above a set threshold to require approval, so that erroneous discounts do not go live and harm margins.
- **US-004** As a Product Owner, I want visibility into catalog data completeness metrics, so that I can prioritize enrichment efforts for underperforming categories.
- **US-005** As a customer browsing in the UAE, I want to see product descriptions and prices in Arabic and AED, so that I can shop comfortably in my local language and currency.
- **US-006** As a Taxonomy Governance Lead, I want to review and approve new category creation requests, so that the category structure remains consistent and does not break PLP navigation.
- **US-007** As a QA Lead, I want an audit trail of all pricing and publish/unpublish actions, so that I can verify compliance during testing and audits.
- **US-008** As a Catalog & Merchandising analyst, I want to be blocked from publishing an incomplete product, so that customers never see a PDP missing images or specifications.

10. Acceptance Criteria

AC ID	Related ID	Criteria	Expected Result
AC-001	FR-001	Given a valid product form, when a user submits title, category, and base attributes, then the product is created in Draft status.	Product record created in Draft status
AC-002	FR-003	Given an existing SKU code, when a user attempts to create a new SKU with the same code, then the system rejects the action.	Duplicate SKU creation blocked with error message
AC-003	FR-008	Given a scheduled SAP feed file, when the import job runs, then valid records are created/updated and invalid records are logged to an exception report.	Successful records imported; exceptions logged
AC-004	FR-012	Given a price change exceeding 15% variance, when the user submits it, then the change is routed to an approver and not applied until approved.	Price held in "Pending Approval" state
AC-005	FR-015	Given a product missing a mandatory image field, when a user attempts to publish, then the system blocks publishing and displays the missing fields.	Publish action blocked with validation message
AC-006	FR-016	Given a product in Active status, when a user unpublishes it, then it is removed from PLP/PDP and the Elasticsearch index within 5 minutes.	Product no longer visible to customers within SLA
AC-007	US-005	Given a customer session set to UAE/Arabic locale, when the customer views a PDP, then description and price display in Arabic and AED.	Correct localized content and currency displayed
AC-008	FR-007	Given a request to create a new second-level category, when submitted, then it is routed to the Taxonomy Governance Lead for approval before going live.	Category remains in "Pending" until approved
AC-009	NFR-002	Given a bulk import file of 50,000 records, when processed, then the job completes within 30 minutes.	Job completion time within SLA

11. Business Rules

- **RULE-001** Every SKU code must be unique across the entire catalog, regardless of market or category.

- **RULE-002** A product cannot transition to "Active" status unless it has at least one image, a description, an assigned category, and a valid price.
- **RULE-003** Price changes exceeding a 15% variance from the current live price require approval from a designated Pricing Manager before going live.
- **RULE-004** New top-level or second-level categories require Taxonomy Governance Lead approval before publication.
- **RULE-005** A product marked "Discontinued" cannot be reactivated directly; it must be recreated as a new product record.
- **RULE-006** All bulk-imported records failing mandatory field validation must be logged to an exception report and excluded from the live catalog until corrected.
- **RULE-007** Pricing for a given SKU must be maintained separately per market currency (INR, AED, and applicable Southeast Asia currencies); no automatic currency conversion is permitted for published prices.
- **RULE-008** Only users with the "Catalog Publisher" role may execute publish/unpublish actions on a product.

12. Process Flow Description

Primary Flow — New Product Onboarding via Bulk Import to Publish:

1. SAP S/4HANA generates a scheduled product master feed file containing new and updated product records.
2. The PCM bulk import service ingests the feed and validates each record against mandatory field and SKU uniqueness rules.
3. Valid records are created or updated in the catalog database in "Draft" status; invalid records are written to an exception report and routed to the Catalog & Merchandising team.
4. A Catalog & Merchandising analyst enriches each draft product with images, descriptions, specifications, and localized content for target markets.
5. The analyst configures pricing per SKU and market currency; changes exceeding the approval threshold are routed to the Pricing Manager.
6. Once pricing is approved (or within auto-approval threshold), the system validates data completeness against publish rules.
7. An authorized Catalog Publisher reviews the enriched product and executes the publish action, setting status to "Active".
8. The publish event triggers a real-time update to the Elasticsearch search index and a stock linkage check against the Inventory Management System (IMS).
9. The product becomes visible on PLP/PDP across web and mobile channels within the defined SLA.

Alternate/Exception Flow — Import Validation Failure:

1. During bulk import validation, a record fails a mandatory field or SKU uniqueness check.
2. The system excludes the record from catalog creation and logs it, with a reason code, to the exception report.
3. The Catalog & Merchandising analyst reviews the exception report and corrects the source data or the SAP product master record.
4. The corrected record is resubmitted for import, either individually or in the next scheduled batch, without requiring reprocessing of the entire original batch.

13. Assumptions

- SAP S/4HANA will remain the authoritative system for base product master and cost data.
- The Inventory Management System (IMS) will expose a stable API for real-time ATP stock linkage per SKU.
- Elasticsearch infrastructure will be scaled to accommodate the target catalog size of 5 million SKUs.
- Business users involved in taxonomy governance and pricing approval will be trained before go-live.
- Multi-language content will be provided by regional content teams; the system will store but not auto-translate content.
- Existing catalog data will be migrated and cleansed prior to cutover to the modernized PCM module.

14. Constraints

- The project must align with the overall ShopSphere Platform Modernization Program (FY2026) timeline and release calendar.

- Category taxonomy changes must not disrupt existing PLP navigation or active promotional campaigns during peak sale periods.
- Bulk import processing windows must avoid conflict with the nightly SAP financial close batch jobs.
- Regulatory requirements in the UAE and Southeast Asia markets regarding product disclosure and pricing transparency must be accommodated within existing field structures.

15. Dependencies

- SAP S/4HANA product master module for base product and cost data feed.
- Inventory Management System (IMS) and NetSuite WMS for stock and warehouse linkage per SKU.
- Elasticsearch cluster availability and indexing pipeline for PLP/PDP search.
- Salesforce CRM for downstream consumption of product data used in personalized recommendations.
- UX Lead deliverables for PDP/PLP content layout standards to inform mandatory field definitions.

16. Risks

Risk ID	Description	Likelihood	Impact	Mitigation
RISK-001	Incomplete or inconsistent SAP product master data causes high bulk import exception rates.	Medium	High	Establish data quality validation rules at SAP source and pre-import staging checks.
RISK-002	Category taxonomy changes made without governance approval break PLP navigation.	Medium	High	Enforce mandatory approval workflow for top/second-level category changes.
RISK-003	Unauthorized pricing changes go live without approval due to workflow bypass.	Low	High	Implement system-enforced approval gate with full audit logging.
RISK-004	Elasticsearch indexing delays cause newly published products to be undiscoverable for extended periods.	Medium	Medium	Implement indexing monitoring and alerting with automated retry.
RISK-005	Multi-language content gaps result in poor customer experience in UAE and Southeast Asia markets.	Medium	Medium	Define minimum localized content requirements per market as a publish gate.
RISK-006	Peak sale event traffic (10x normal) causes catalog search performance degradation.	Low	High	Load-test Elasticsearch and API tier ahead of major sale events; enable auto-scaling.

17. Data Requirements

The PCM module captures and stores product, category, pricing, and content data required to render PLP/PDP experiences and to support downstream analytics and personalization. PII exposure is minimal within this module, as catalog data is largely non-personal product metadata; however, internal user actions (e.g. who approved a price change) are logged with employee identifiers and must be protected per internal access control policy. Catalog data shall be retained indefinitely for active and inactive products to preserve historical reporting, while discontinued product records shall be retained for a minimum of 7 years to satisfy financial and regulatory audit requirements linked to SAP S/4HANA transaction history.

Data Entity	Key Fields	Notes
Product	Product ID, Title, Brand, Category, Status, Description	Synced from SAP S/4HANA; multi-language fields supported
SKU/Variant	SKU Code, Size, Color, Product ID, ATP Quantity	SKU Code globally unique; linked to IMS for stock
Category	Category ID, Name, Parent Category, Attribute Set	Hierarchical, up to 5 levels
Price	SKU Code, Market Currency, Base Price, Promo Price, Approval Status	Maintained per market currency; audit trail required
Content Asset	Product ID, Image URLs, Specification Fields	Minimum one image required for publish

18. Integration Requirements

System	Purpose	Direction	Data Exchanged
SAP S/4HANA (ERP)	Product master synchronization and cost data feed	Inbound to PCM	Product ID, title, brand, base cost, category mapping

Inventory Management System (IMS) / NetSuite WMS	Stock linkage and ATP quantity per SKU	Bidirectional	SKU code, warehouse stock levels, ATP status
Elasticsearch	Catalog search indexing for PLP/PDP	Outbound from PCM	Product, SKU, category, price, and content fields for indexing
Salesforce CRM	Feed product data for personalized recommendations	Outbound from PCM	Product ID, category, price, attributes

19. Reporting Requirements

- Catalog data completeness dashboard by category, showing percentage of products meeting mandatory PDP field standards.
- Bulk import exception report, detailing rejected records, reasons, and resolution status.
- Pricing approval audit report, listing all pending, approved, and rejected price change requests.
- Time-to-publish report, tracking average duration from product creation to "Active" status.
- Category taxonomy change log, showing all category creation/modification requests and approval status.
- Search indexing latency report, monitoring the time between publish action and Elasticsearch index update.

20. Success Metrics

Metric	Target
Catalog data completeness rate (PDP mandatory fields)	≥ 98% of active products
Average time-to-publish (draft to active)	≤ 24 hours for standard products
Bulk import exception rate	< 3% of records per batch
Search index update latency post-publish	≤ 5 minutes
Unauthorized pricing incidents	Zero incidents per quarter
Catalog-related customer complaints	Reduction of 30% within 2 quarters post-launch

21. Future Enhancements

- AI-based dynamic repricing engine leveraging demand and competitor signals (explicitly out of scope for this phase).
- Standalone digital asset management (DAM) system for centralized image and media governance.
- Marketplace third-party seller self-service catalog onboarding portal.
- Automated AI-assisted content enrichment (description generation, image quality scoring).
- Expanded multi-language support beyond the initial four languages to cover additional Southeast Asia markets.

22. Glossary

Term	Definition
SKU	Stock Keeping Unit — a unique identifier for a specific product variant.
PDP	Product Detail Page — the customer-facing page showing full product information.
PLP	Product Listing Page — the customer-facing page listing multiple products in a category or search result.
ATP	Available to Promise — the quantity of stock available for sale and fulfillment commitment.
GMV	Gross Merchandise Value — total value of goods sold through the platform.
B2C	Business-to-Consumer — a business model selling directly to end consumers.
API	Application Programming Interface — a defined interface for system-to-system data exchange.
PII	Personally Identifiable Information — data that can identify an individual.
SLA	Service Level Agreement — a defined commitment on service performance or turnaround.
Taxonomy	The hierarchical category structure used to organize products in the catalog.

23. Appendix

Reference Documents: ShopSphere Platform Modernization Program Charter (FY2026); SAP S/4HANA Product Master Integration Specification; Elasticsearch Indexing Design Document.

Related BRDs: BRD-EC-ORD-001 Order Management Module; BRD-EC-INV-003 Inventory & Warehouse Management Module; BRD-EC-PAY-004 Payment Gateway Integration Module.

Version	Date	Author	Change Description
0.1	05-Jan-2026	Neha Kapoor	Initial draft circulated for stakeholder review
1.0	20-Jan-2026	Neha Kapoor	Incorporated feedback from Product, IT Architecture, and Compliance; approved by Ananya Sharma